Referee's report on the paper

Jungkai A. Chen, Christopher D. Hacon: On Ueno's conjecture K

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Ueno's conjecture K concerns smooth projective varieties with Kodaira dimension zero. In short, it is supposed to provide information on the structure of the Albanese fibration of such a variety. Of its predictions, Kawamata has established that the Albanese fibration is an algebraic fibre space, but the conjectures on the Kodaira dimension of a general fibre and the structure theorem for X remain open; these are not settled by the present article either.

Earlier work of Kawamata regarding a more general set of conjectures coming from Iitaka shows that in fact Ueno's conjecture K follows from the full minimal model program including abundance. Although there has been considerable progress on the minimal model program recently, due mainly to Hacon (the second author of the present work) and McKernan, the abundance conjecture seems to be way out of reach at the moment. This way, a positive result in the direction of Ueno's conjecture K can also be considered as a (somewhat remote, but nevertheless valid and useful) reality check for abundance.

More concretely, a part of Ueno's conjecture is the numerical prediction that for a smooth projective variety with $\kappa(X) = 0$, and Albanese morphism $a : X \to A$, one has

$$\kappa(F) = 0$$

for a general fibre F. The main result of Chen and Hacon goes in this direction, namely, they establish that under the abovementioned circumstances

$$\kappa(F) \leq \dim A$$

holds.

I believe that the result has a place in an outstanding journal like the Mathematische Annalen, whence, *I suggest that the paper be accepted for publication*. However, I need to add that I consider the article to be borderline. The reason for this is on the one hand that the methods have already been used by the authors for similar purposes many times in the past; on the other hand, the actual result achieved does not come close to the goal mentioned in the title. In any case, I think that the technical complexity of the proof sheds enough light on the difficulty of the problem at hand, which justifies the publication.

Just as in the case of many similar results in this area (pioneered in large part by the authors), the proof comes as an elegant application of the theory of the Fourier–Mukai transforms, relying on ideas coming from the original paper of Mukai. The theory enters in the form of a bijection given by the Fourier–Mukai transform between the category of unipotent vector bundles on an abelian variety A and the category of Artinian $\mathcal{O}_{A,0}$-modules of finite length.

The other crucial ingredient of the verification is an involved computation with Artinian $\mathcal{O}_{A,0}$-modules; in a way this is the actual main novelty of the paper. These calculations are sometimes a little bit tedious to follow; some improvements in the exposition would be welcome.

Some other decidedly minor issues that could be addressed: first, the word 'generated' is often used in place of the standard terminology that would be 'globally generated'. I would suggest sticking to the established usage. Second, although this is admittedly a question of personal taste, the fact that the Introduction contains displayed results of various people beside that of the authors' is somewhat disorienting.